Aman Haris

Bengaluru, Karnataka, India | amanharisofficial@protonmail.com | +91-8618985209 | linkedin.com/in/aman-haris

https://aman-haris-portfolio.onrender.com

About Me

Software Engineer with expertise in **AI, GenAI, ML** and **Automation**. Proven track record in developing impactful solutions and improving business efficiency through advanced AI technologies. Seeking new challenges in AI and ML to drive innovation.

Key Skills

AI/ML Technologies: Generative AI, Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), Natural Language Processing, Computer Vision

Programming & Development: Python, SQL, Postgres, HTML, CSS, JS

Cloud & DevOps: Azure AI Services, Azure App Services, Azure Container Registry, Azure Storage, Azure Entra ID, Azure Repos, Azure MySQL, Azure Function Apps, MSAL / Azure AD Auth, Docker, Nginx, AWS S3, AWS SQS, AWS Bedrock, AWS Bedrock Agentcore, AWS RDS, AWS Lambda, AWS SNS, AWS Comprehend

Frameworks & Libraries: Langchain, FastAPI, React, SQLAlchemy, Huggingface, OpenAI, Pinecone, Streamlit,

Tools & Platforms: Anaconda, VS Code, Postman, PowerBI, BluePrism, SQL Workbench, Figma, Git, DBeaver

Professional Skills: Technical Leadership & Mentoring, Process Implementation & Optimization, Cross-Functional Collaboration, Project Management & Ownership, Continuous Learning & Innovation

Experience

Software Engineer 1 - Tessolve Semiconductors Pvt Ltd — Feb 2024 – April 2026
Software Engineer 2 - Tessolve Semiconductors Pvt Ltd — May 2026 – Present

Key Projects:

- **Resume Analysis Tool (ATS) — Actively Maintained:** Built and continue to enhance an AI-powered ATS platform that scores resumes against job descriptions with a deterministic (non-LLM-scored) JD-match algorithm, backed by an async AWS SQS worker queue with live SSE progress tracking and S3-persisted sessions. So far, 70k+ resumes have been analyzed by the tool, improving recruitment efficiency by 60%
- **SchematicIQ (Schematic Verification Tool) — Actively Maintained:** Engineered a tester-board validation tool that traces relay/signal paths via BFS graph traversal and cross-checks netlists against resource-allocation sheets, with direct-to-S3 presigned uploads, reducing manual wiring verification effort by 75%
- **Tessa Embedded Bot & Analytics Dashboard — Actively Maintained:** Designed and deployed an AI front-desk assistant for the Embedded micro-website (RAG Q&A, guided service requests, career applications) that met requirements the vendor could not, paired with a companion analytics dashboard using AWS Comprehend sentiment analysis for lead and engagement tracking. Live at embedded.tessolve.com
- **Platform Modernization & Security:** Led migration of the team's production AI tools (Assertify, Resume Analysis Tool, RAG Chatbot, Sales Suite) from Streamlit prototypes to a scalable React + FastAPI architecture with centralized logging; hardened endpoints with MSAL-based authentication and introduced Shannon AI Pentester for automated vulnerability and prompt-injection testing, contributing a fix upstream (KeygraphHQ/shannon#151)
- **Assertify:** Developed an innovative LLM-powered tool that automates the conversion between English and SystemVerilog Assertions, reducing manual verification effort by 50% and improving code quality
- **Sales Suite:** Built a suite of GenAI agents to support lead capture, and a biweekly Sales Mail Automation script that generates company insights for sales reps to improve outreach efficiency
- **RAG-based Chatbot:** Architected and deployed a chatbot system using Langchain, OpenAI, Azure, and Pinecone, enhancing cross-departmental information retrieval by 85%
- **ASPICE Studio:** Engineered a GenAI tool to assist automotive embedded engineers to generate variouse documentation artifacts such as interface tables, flowcharts, uml diagrams etc from the Automotive C codebase in the SWE 3 and SWE2 phase of ASPICE. The tool helped reduce the development time by 90%.
- **Veriplan:** Developed a RAG tool that streamlines verification planning across various levels with guided workflows and intelligent prompts, enabling DV Engineers to create test plans and manage comprehensive coverage with minimal manual effort.

Leadership & Process Improvement:

- Authored Testing SOPs and led adoption of Software Development Lifecycle (SDL) and CI/CD processes for the AI team, establishing standardized workflows that improved delivery quality and reduced rework.
- Implemented comprehensive Azure cloud architecture for AI applications, ensuring scalability, security, and optimal performance through containerization and CI/CD pipelines
- Proactively governed cloud costs and model lifecycle — cleaned up unused Azure resources and upgraded legacy LLM deployments for better accuracy and cost-efficiency.
- Established best practices for code modularity, documentation, and testing, resulting in improved maintenance efficiency and knowledge transfer across the engineering team.
- Provided mentorship to new team members and interns, accelerating their integration and productivity.
- Collaborated across multiple business units, including VLSI, Embedded, Sales, Test, Legal, HR, etc, to develop tailored AI solutions that address specific departmental challenges.
- Consistently rated **Excellent/Outstanding** across all review cycles, including an **Outstanding Performance** rating for "exceeding expectations by a huge stretch, going above and beyond immediate areas of responsibility."

Education

MSc Artificial Intelligence, Heriot-Watt University Edinburgh — Sept 2022 – Dec 2023

BE in Computer Science, AMC Engineering College Bengaluru — Aug 2018 – Aug 2022

Languages

English (Professional), Malayalam (Native), Kannada (Fluent), Hindi (Fluent), Tamil (Conversational)